

MATH1231 CALCULUS

Chapter 4B

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Sequences

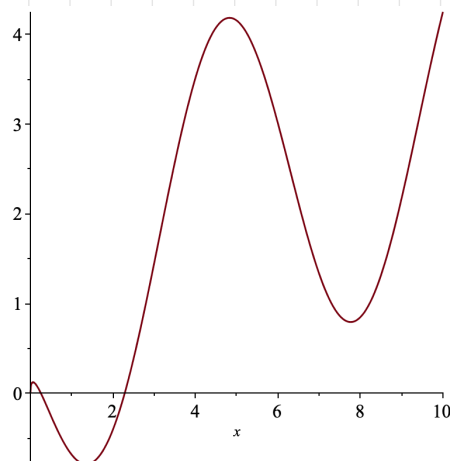
(4.3) Sequences

Sequences arise naturally in all areas of mathematics and science, not just in the context of Taylor polynomials.

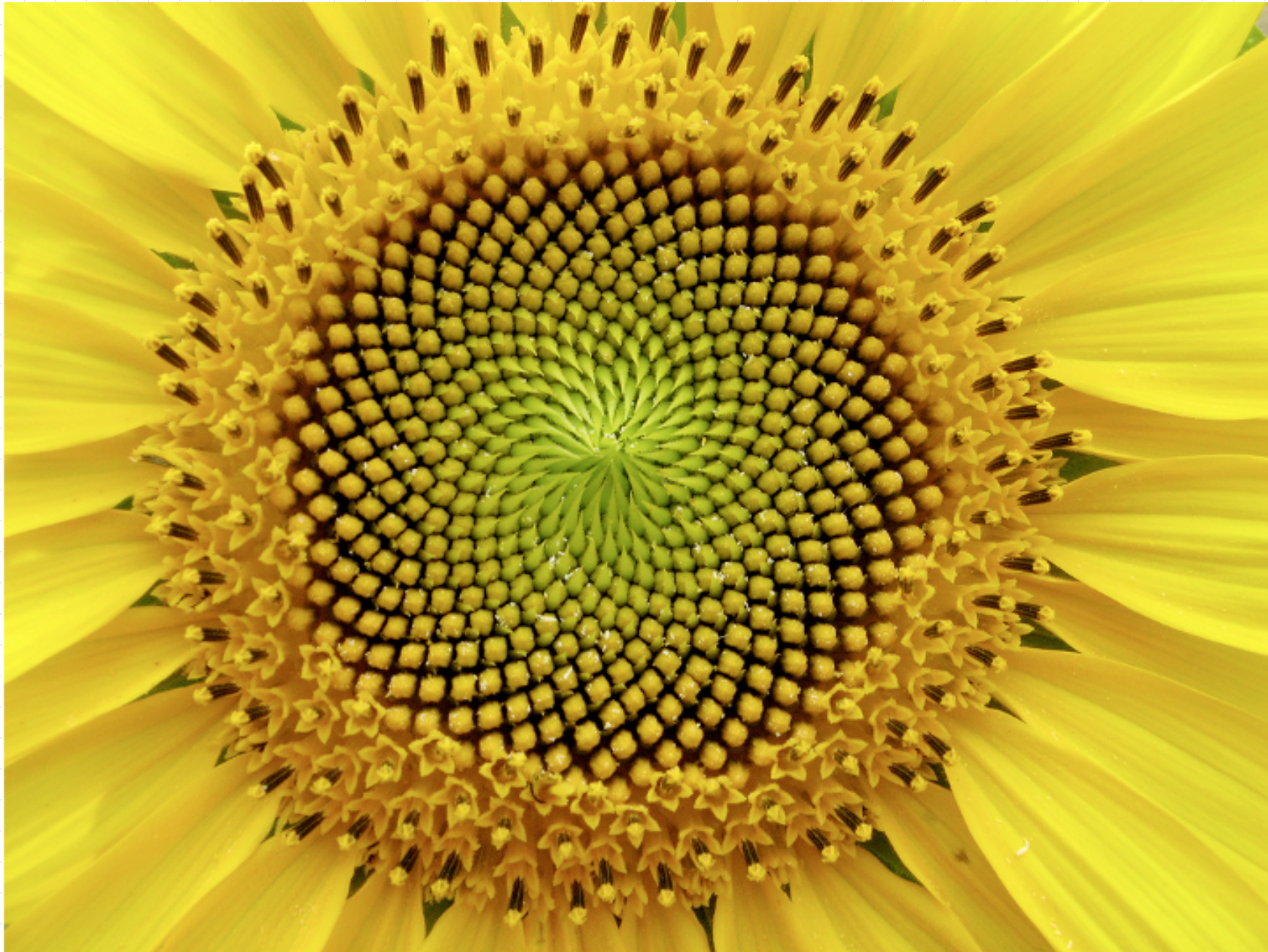
Sequences often arise when a process is discrete rather than continuous, or when we solve nonlinear equations iteratively (e.g., Newton's method).

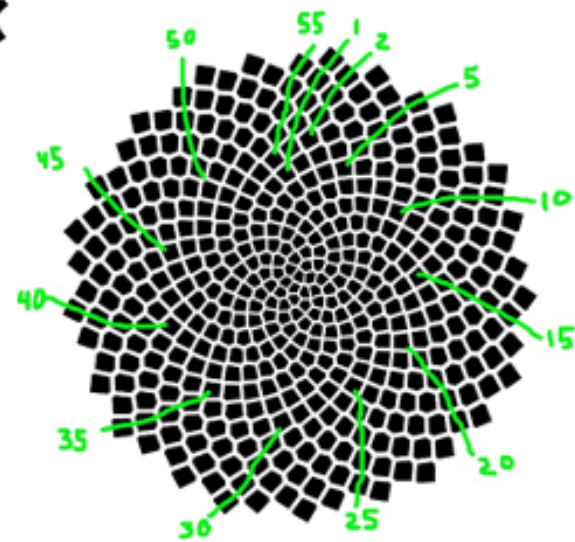
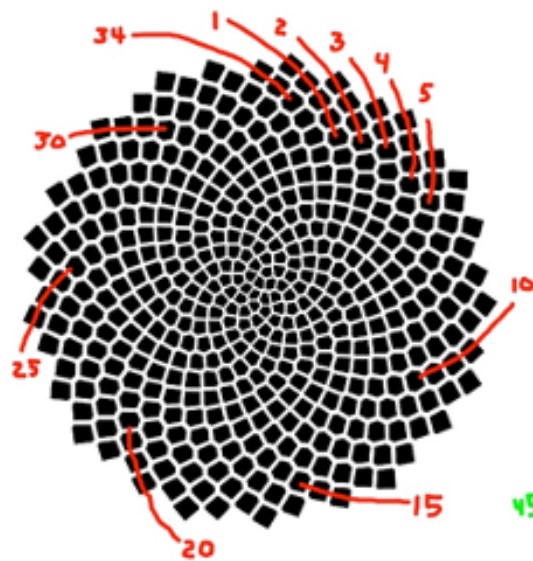
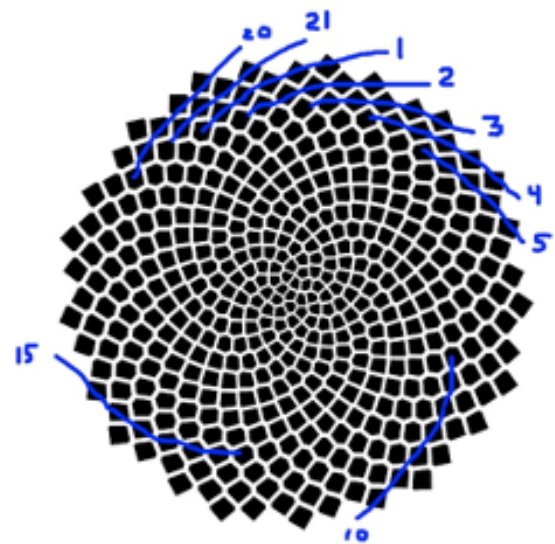
For example, how can we find the real roots of the function

$$f(x) = \sqrt{x} - 2 \sin x?$$



$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)}, \quad \text{1d Newton's method}$$





Fibonacci Sequence

$$\{F_n\} = \{1, 1, 2, 3, 5, 8, 13, 21, \dots\}$$

$$F_{n+1} = F_n + F_{n-1}, \quad \text{with} \quad F_0 = 1, F_1 = 1$$

The Fibonacci numbers grow exponentially and $\{F_n\}$ is a *divergent* sequence. Consider $a_n = F_{n+1}/F_n$:

$$\{a_n\} = \left\{ \frac{1}{1}, \frac{2}{1}, \frac{3}{2}, \frac{5}{3}, \frac{8}{5}, \frac{13}{8}, \frac{21}{13}, \dots \right\}$$

The sequence $\{a_n\}$ is a *convergent* sequence with the limit

$$\lim_{n \rightarrow \infty} a_n = \frac{1}{2} (1 + \sqrt{5}) \sim 1.618$$

the **Golden Mean**

What is a Sequence?

A sequence is a **function** f whose domain is $\mathbb{N}$.
The codomain is $\mathbb{R}$, i.e., $f : \mathbb{N} \rightarrow \mathbb{R}$

Notation

- ▶ Usually use a and b rather than f and g .
- ▶ Usually write a_n instead of $a(n)$.
- ▶ Usually denote a sequence by $\{a_n\}_{n=0}^{\infty}$ or $\{a_n\}$.
- ▶ A sequence does not have to start at $n = 0$;
e.g., $\{a_n\}_{n \geq 1}$
- ▶ The order of the terms in a sequence is important.
 $\{1, 2, 3, 4, \dots\}$ is *not* the same as $\{3, 1, 2, 4, \dots\}$

Examples – closed form

1. $a_n = n^2$ for $n = 0, 1, 2, \dots$

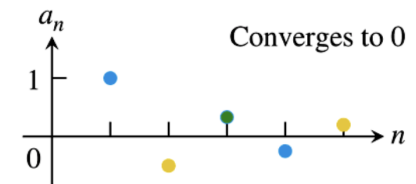
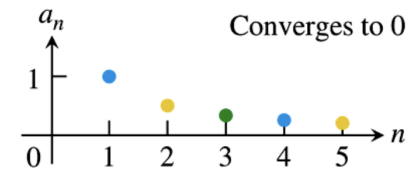
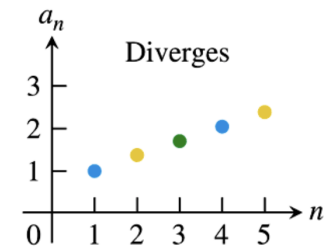
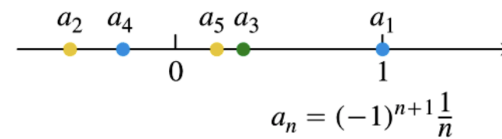
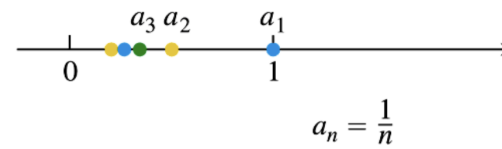
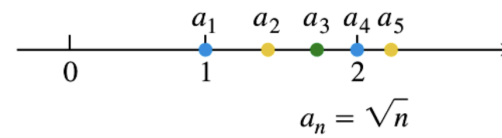
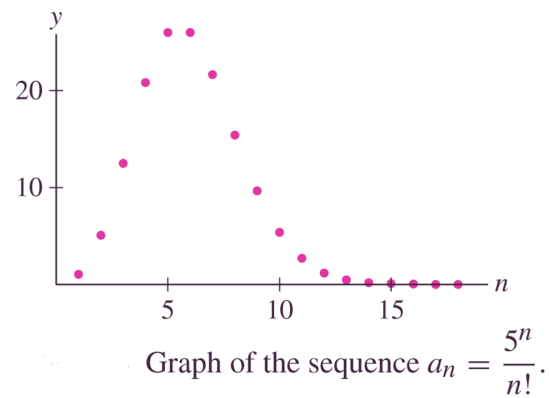
2. $b_n = (-1)^n$

3. $c_n = \frac{1}{n+1}$

4. $d_n = 2^n$

Or we can list the terms of the sequence:

Graphs of sequences



(4.3.1) Convergent sequences

What happens in the sequence $\{a_n\}$ as $n \rightarrow \infty$?

Does it approach a limit $L \in \mathbb{R}$?

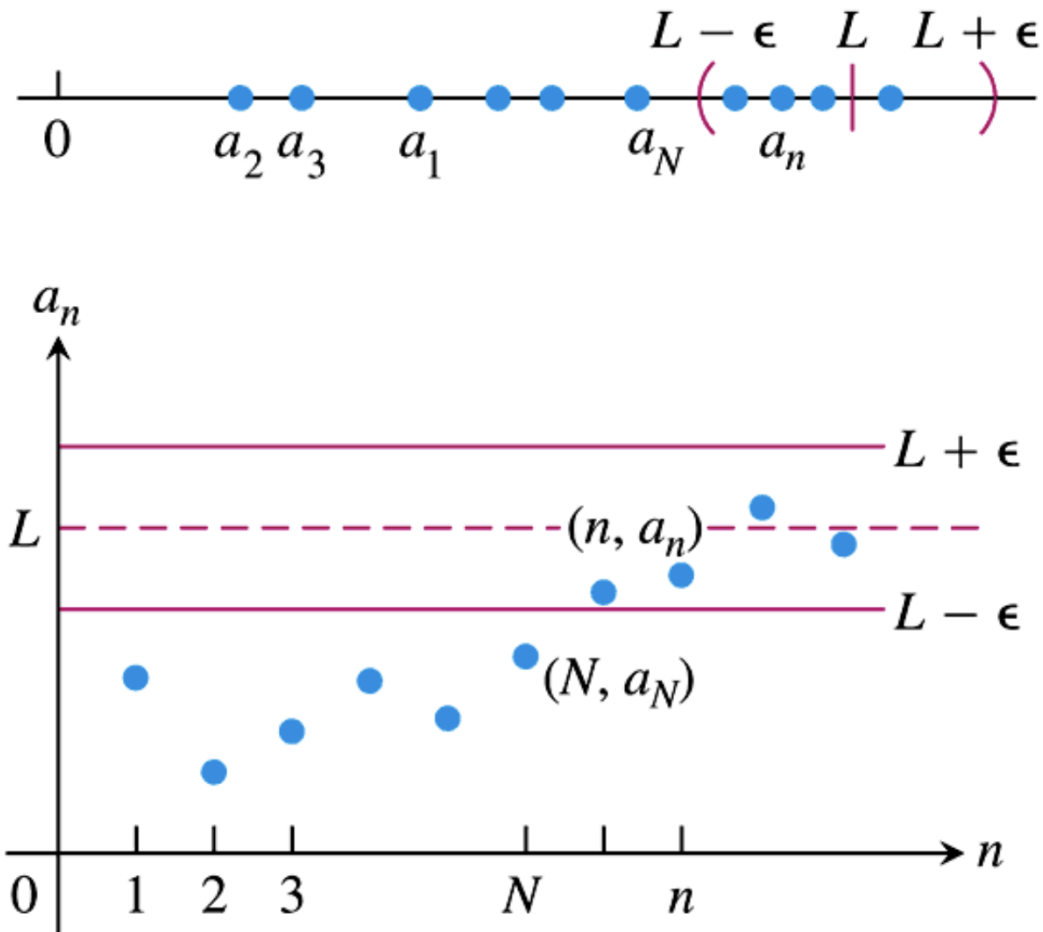
Definition

L is the **limit** of $\{a_n\}$ if for every $\epsilon > 0$ there is $N \in \mathbb{N}$ such that $|a_n - L| < \epsilon$ whenever $n \geq N$.

Notation

$$\lim_{n \rightarrow \infty} a_n = L \quad \text{or} \quad a_n \rightarrow L \quad \text{as} \quad n \rightarrow \infty$$

The formal limit graphically



Example

Prove formally that $\lim_{n \rightarrow \infty} \frac{1}{n} = 0$.

Example

Prove formally that $\lim_{n \rightarrow \infty} \frac{3n^2 + 7}{n^2 + 2} = 3$.

Divergent Sequences

Divergent sequences can be classified as follows

- ▶ (i) a_n **diverges to** ∞ $a_n \rightarrow \infty$ as $n \rightarrow \infty$.
- ▶ (ii) a_n **diverges to** $-\infty$ $a_n \rightarrow -\infty$ as $n \rightarrow \infty$.
- ▶ (iii) $\lim a_n$ does not exist

Examples:

1. $a_n = n^2$

2. $a_n = -n$

3. $a_n = (-1)^n$

(4.3.2) Basic theorems on limits

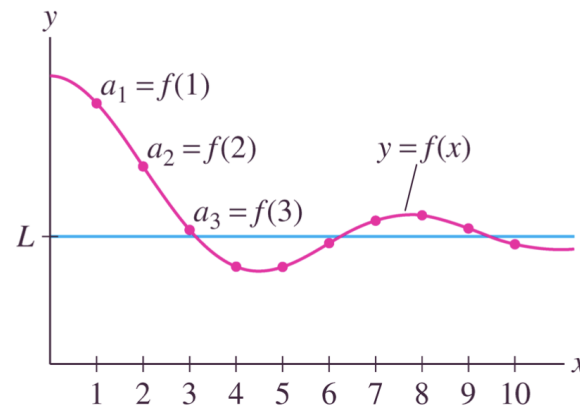
Suppose $a_n \rightarrow A$ and $b_n \rightarrow B$ as $n \rightarrow \infty$ then

Rule 1 $a_n + b_n \rightarrow A + B$ unless $A + B$ has the form $\infty - \infty$

Rule 2 $a_n b_n \rightarrow AB$ unless AB has the form $0 \cdot \pm\infty$

Rule 3 $\frac{a_n}{b_n} \rightarrow \frac{A}{B}$ unless $\frac{A}{B}$ has the form $\frac{0}{0}$ or $\frac{\infty}{\infty}$

Rule 4 Given $a_n = f(n)$ for $n \in \mathbb{N}$ define a function $f(x)$ for $x \in \mathbb{R}$.
Then $\lim_{x \rightarrow \infty} f(x) = L \Rightarrow \lim_{n \rightarrow \infty} f(n) \rightarrow L$.



If $f(x)$ converges to L , then the sequence $a_n = f(n)$ also converges to L .

Continuous function Theorem for Sequences

IF

- i) $\{a_n\}$ is a sequence with $\lim_{n \rightarrow \infty} a_n = A$
- ii) $f(x)$ is continuous at A , for $x \in \mathbb{R}$
- iii) a_n lies in the domain of f for all n

THEN

$$\lim_{n \rightarrow \infty} f(a_n) = f\left(\lim_{n \rightarrow \infty} a_n\right)$$

Examples



$$\lim_{n \rightarrow \infty} \frac{n+4}{n+1}$$



$$\lim_{n \rightarrow \infty} \left(\frac{n+4}{n+1} \right)^n$$

L'Hopital's Rule

Given $f(n)$ and $g(n)$ for $n \in \mathbb{N}$ define $f(x)$ and $g(x)$ for $x \in \mathbb{R}$.

If $f(x)$ and $g(x)$ are differentiable and:

i)

if $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$

or if $f(x) \rightarrow \infty$ and $g(x) \rightarrow \infty$

ii)

$$\frac{f'(x)}{g'(x)} \rightarrow L \in \mathbb{R} \quad \text{as } x \rightarrow \infty$$

then

$$\lim_{n \rightarrow \infty} \frac{f(n)}{g(n)} \rightarrow L$$

Example

Find $\lim_{n \rightarrow \infty} \frac{n^2 - 2n + 1}{2n^2 + 5}.$

Example

Find $\lim_{n \rightarrow \infty} \sin\left(\frac{\pi}{2} - \frac{1}{n}\right).$

Example

Find $\lim_{n \rightarrow \infty} \sqrt{n^2 + 3n} - n$

Example

Find $\lim_{n \rightarrow \infty} n \sin \frac{1}{n}$

Common Limits

$$\lim_{n \rightarrow \infty} \frac{\ln n}{n} = 0 \quad (1)$$

$$\lim_{n \rightarrow \infty} \sqrt[n]{n} = 1 \quad (2)$$

$$\lim_{n \rightarrow \infty} x^{1/n} = 1 \quad (x > 0) \quad (3)$$

$$\lim_{n \rightarrow \infty} x^n = 0 \quad (|x| < 1) \quad (4)$$

$$\lim_{n \rightarrow \infty} \left(1 + \frac{x}{n}\right)^n = e^x \quad (\text{any } x) \quad (5)$$

$$\lim_{n \rightarrow \infty} \frac{x^n}{n!} = 0 \quad (\text{any } x) \quad (6)$$

Growth of sequences with n

Who's the biggest of them all?

a_n	growth as $n \rightarrow \infty$
1	constant
$\ln n$	slow growth
$n^k \quad (k > 0)$	faster for larger k
$c^n \quad (c > 1)$	faster for larger c
$n!$	rapid
n^n	very rapid

Pinching Theorem for Sequences

(also called “Squeezing Thm.”, “Sandwich Thm.”)

Suppose that $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ are sequences such that for all sufficiently large n we have

$$a_n \leq b_n \leq c_n.$$

If $\lim_{n \rightarrow \infty} a_n = L = \lim_{n \rightarrow \infty} c_n$ then $\lim_{n \rightarrow \infty} b_n$ exists and $\lim_{n \rightarrow \infty} b_n = L$.

Example

Show that $\lim_{n \rightarrow \infty} \frac{(-1)^n}{n} = 0$

Example

Find

$$\lim_{n \rightarrow \infty} \frac{n!}{n^n}$$

Bounded monotonic sequences converge

Theorem:

- ▶ I) Let a_n be a non-decreasing sequence that is bounded above, i.e.,

$$a_0 \leq a_1 \leq \dots \leq a_{n-1} \leq a_n \leq \dots \leq K$$

Then a_n converges to some limit $L \leq K$

- ▶ II) Let a_n be a non-increasing sequence that is bounded below, i.e.,

$$a_0 \geq a_1 \geq \dots \geq a_{n-1} \geq a_n \geq \dots \geq K$$

Then a_n converges to some limit $L \geq K$

Example

Suppose we are trying to find the positive root of $x^2 = x + 1$ by generating the iteratively defined sequence

$$a_{n+1} = \sqrt{a_n + 1}, \quad a_0 = 1$$

Show the $\lim_{n \rightarrow \infty} a_n$ exists.

Proof ingredients: 1.) is $\{a_n\}$ monotonic? 2.) is $\{a_n\}$ bounded?